

Final Project Report ECE-111

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What is SHA 256 and Bitcoin Hashing?

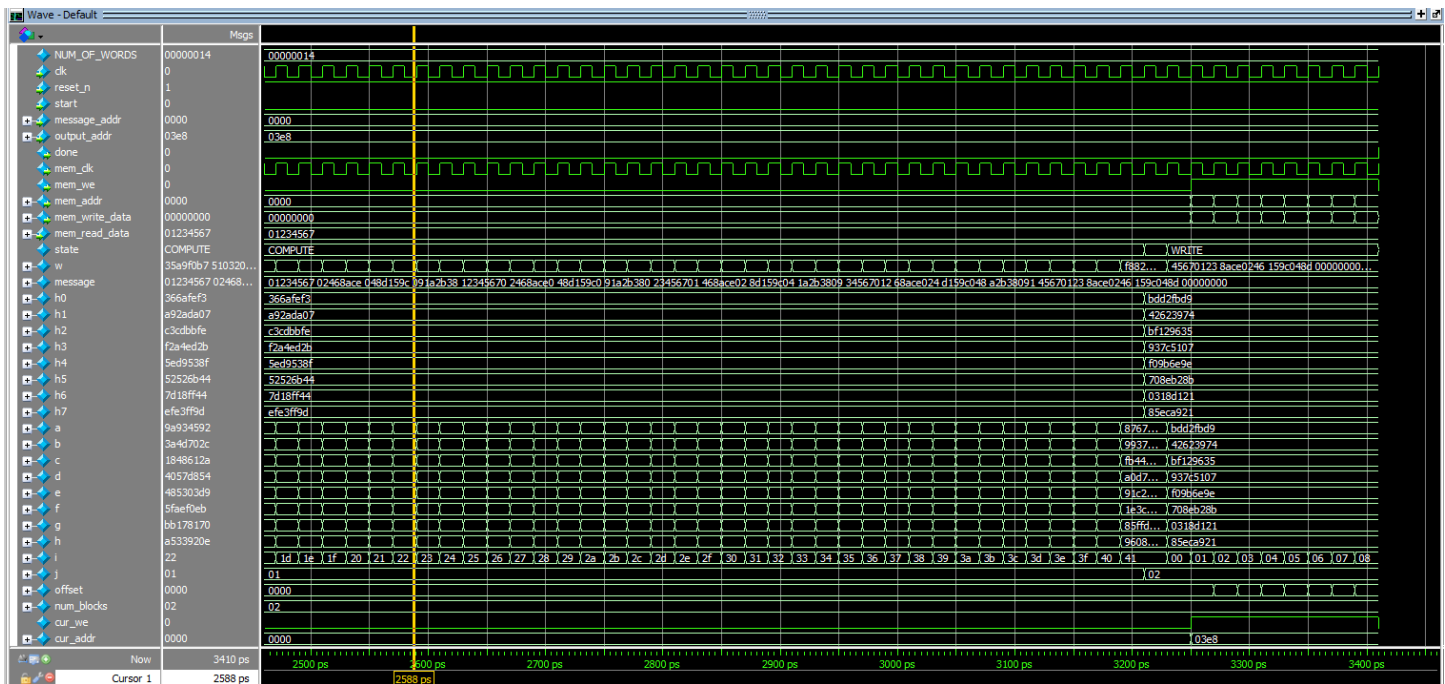
SHA-256 and bitcoin hashing are both hash algorithms used to create a code to check if the data integrity is violated during the process of transferring or to save and compare password during authentication process. In the case of Bitcoin, since all block are connected using this algorithm into a chain (blockchain), whenever there's a slight change to any part of the chain, every value will change accordingly, giving all the owner to identify the integrity of each block data and make sure that there's no change to the data itself through cross checking of different block, checking if a malicious user has modified the block.

What is the Algorithm?

The algorithm uses constant right rotation of the value and message extension to output the hashed value in the same length. Each bit is grouped and processed multiple times using right rotation and combined with bits around, which ultimately creates a hash valued that can't reversely calculate the original value, however, the hash value will always be the same for each input, giving user the ability to identify if there's change in the original data.

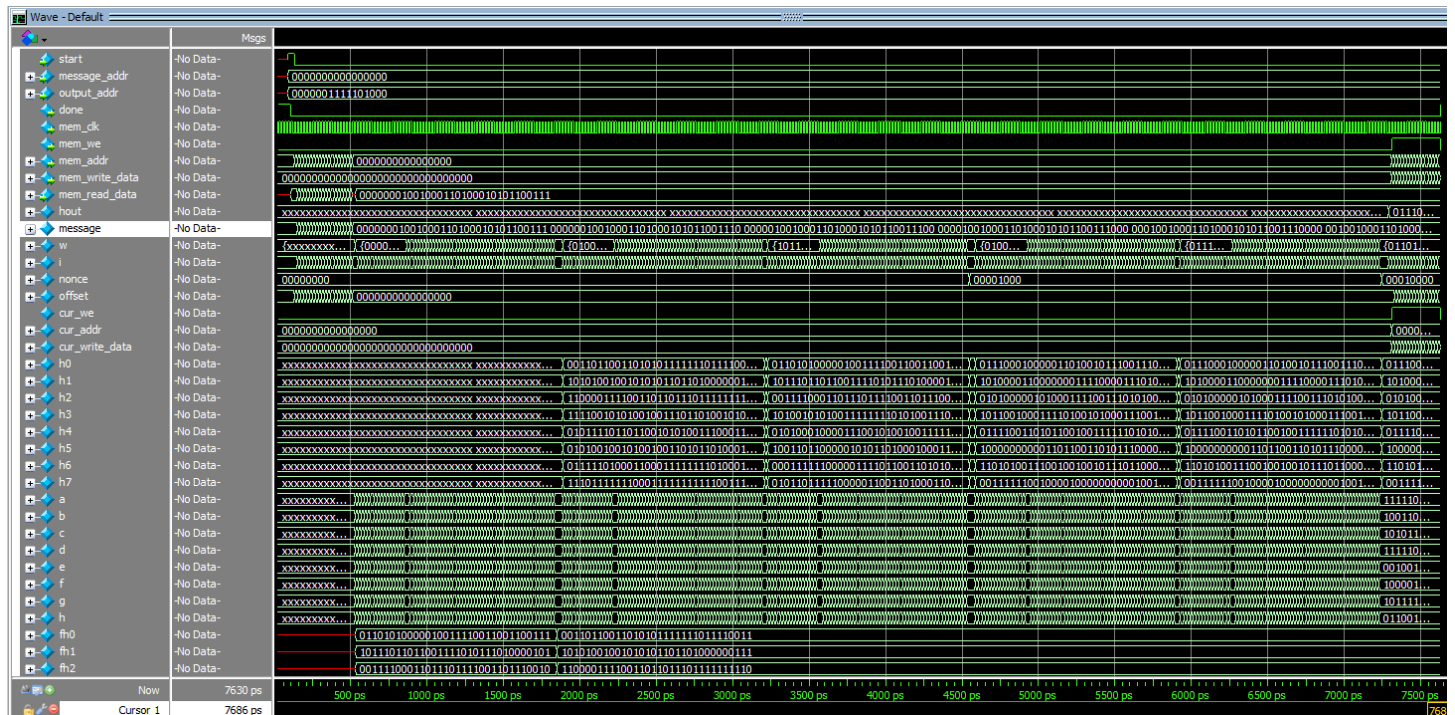
Simulations

simplified_sha256



```
VSIM 2> run -all
#
# -----
# MESSAGE:
#
# -----
# 01234567
# 02468ace
# 048d159c
# 091a2b38
# 12345670
# 2468ace0
# 48d159c0
# 91a2b380
# 23456701
# 468ace02
# 8d159c04
# 1a2b3809
# 34567012
# 68ace024
# d159c048
# a2b38091
# 45670123
# 8ace0246
# 159c048d
# 00000000
# *****
#
# -----
# COMPARE HASH RESULTS:
#
# Correct H[0] = bdd2fbd9 Your H[0] = bdd2fbd9
# Correct H[1] = 42623974 Your H[1] = 42623974
# Correct H[2] = bf129635 Your H[2] = bf129635
# Correct H[3] = 937c5107 Your H[3] = 937c5107
# Correct H[4] = f09b6e9e Your H[4] = f09b6e9e
# Correct H[5] = 708eb28b Your H[5] = 708eb28b
# Correct H[6] = 0318d121 Your H[6] = 0318d121
# Correct H[7] = 85eca921 Your H[7] = 85eca921
# *****
#
# CONGRATULATIONS! All your hash results are correct!
#
# Total number of cycles:      168
#
# *****
#
# ** Note: $stop      : C:/Users/Jaspe/repositories/ECE 1
```

bitcoin_hash



```
VSIM> run -all
# -----
# 19 WORD HEADER:
# -----
# 01234567
# 02468ace
# 048d159c
# 091a2b38
# 12345670
# 2468ace0
# 48d159c0
# 91a2b380
# 23456701
# 468ace02
# 8d159c04
# 1a2b3809
# 34567012
# 68ace024
# d159c048
# a2b38091
# 45670123
# 8ace0246
# 159c048d
# *****
#
# -----
# COMPARE HASH RESULTS:
# -----
# Correct H0[ 0] = 7106973a Your H0[ 0] = 7106973a
# Correct H0[ 1] = 6e66eea7 Your H0[ 1] = 6e66eea7
# Correct H0[ 2] = fbef64dc Your H0[ 2] = fbef64dc
# Correct H0[ 3] = 0888a18c Your H0[ 3] = 0888a18c
# Correct H0[ 4] = 9642d5aa Your H0[ 4] = 9642d5aa
# Correct H0[ 5] = 2ab6af8b Your H0[ 5] = 2ab6af8b
# Correct H0[ 6] = 24259d8c Your H0[ 6] = 24259d8c
# Correct H0[ 7] = ffb9bcd9 Your H0[ 7] = ffb9bcd9
# Correct H0[ 8] = 642138c9 Your H0[ 8] = 642138c9
# Correct H0[ 9] = 054cafc7 Your H0[ 9] = 054cafc7
# Correct H0[10] = 78251a17 Your H0[10] = 78251a17
# Correct H0[11] = af8c8f22 Your H0[11] = af8c8f22
# Correct H0[12] = d7a79ef8 Your H0[12] = d7a79ef8
# Correct H0[13] = c7d10c84 Your H0[13] = c7d10c84
# Correct H0[14] = 9537acfd Your H0[14] = 9537acfd
# Correct H0[15] = c1e4c72b Your H0[15] = c1e4c72b
# *****
#
# CONGRATULATIONS! All your hash results are correct!
#
# Total number of cycles:      379
#
# *****
```

Resource Usage Summary

simplified_sha256

Analysis & Synthesis Resource Usage Summary		
 <<Filter>>		
	Resource	Usage
1	▼ Estimated ALUTs Used	1898
1	-- Combinational ALUTs	1898
2	-- Memory ALUTs	0
3	-- LUT_REGS	0
2	Dedicated logic registers	1751
3		
4	▼ Estimated ALUTs Unavailable	0
1	-- Due to unpartnered combinational logic	0
2	-- Due to Memory ALUTs	0
5		
6	Total combinational functions	1898
7	▼ Combinational ALUT u... by number of inputs	
1	-- 7 input functions	0
2	-- 6 input functions	311
3	-- 5 input functions	4
4	-- 4 input functions	14
5	-- <=3 input functions	1569
8		
9	> Combinational ALUTs by mode	
10		
11	Estimated ALUT/register pairs used	2527
12		
13	> Total registers	1751
14		
15		
16	I/O pins	118
17		
18	DSP block 18-bit elements	0
19		
20	Maximum fan-out node	clk~input
21	Maximum fan-out	1752
22	Total fan-out	13437
23	Average fan-out	3.46

bitcoin_hash

Analysis & Synthesis Resource Usage Summary		
 Filter		
	Resource	Usage
1	▼ Estimated ALUTs Used	17655
1	-- Combinational ALUTs	17655
2	-- Memory ALUTs	0
3	-- LUT_REGS	0
2	Dedicated logic registers	11706
3		
4	▼ Estimated ALUTs Unavailable	2248
1	-- Due to unpartnered combinational logic	2248
2	-- Due to Memory ALUTs	0
5		
6	Total combinational functions	17655
7	▼ Combinational ALUT u... by number of inputs	
1	-- 7 input functions	700
2	-- 6 input functions	2994
3	-- 5 input functions	2191
4	-- 4 input functions	2914
5	-- <=3 input functions	8856
8		
9	> Combinational ALUTs by mode	
10		
11	Estimated ALUT/register pairs used	23501
12		
13	> Total registers	11706
14		
15		
16	I/O pins	118
17		
18	DSP block 18-bit elements	0
19		
20	Maximum fan-out node	clk~input
21	Maximum fan-out	11707
22	Total fan-out	119945
23	Average fan-out	4.05